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A New Model of Occlusion

Biomechanical Principles of Occlusion

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Lightly reformatted; typography corrected from the original scan.

Dentistry has long recognized the importance of occlusion in dental treatment as it impacts patient comfort, durability of teeth and restorations, and the stability of the bite. One's concept of occlusion is vitally important, as it is the overriding structure that guides treatment, whether for orthodontics, TMJ treatment, or crown and bridge.

Bernard Jankelson placed occlusion on an objective scientific basis with the development of neuromuscular concepts and instrumentation in the 1970s. These concepts were a great advancement over Angle's classification in that they (1) measured muscles, (2) utilized TENS to generate a bite without torque, and (3) incorporated the importance of muscles — not just teeth — when assessing proper occlusion.

As neuromuscular bite registration techniques use resting muscles to determine three-dimensional jaw position, errors can occur in determining jaw alignment for various reasons. As a cross-check, the position of the mandible during speech has proved helpful. More recently, clinical experience has shown the “fair arc of opening and closing” to be likewise extremely useful.

Because occlusal concepts have been advanced primarily by prosthetic-oriented dentists, current neuromuscular occlusal theory has not incorporated orthodontic and orthopedic wisdom. European dental orthopedic experience has shown that, through appropriate therapy, mandibular rest position can be advanced. As a practitioner of European mandibular advancement concepts for over twenty-five years, I have long tried to understand the principles involved and to reconcile them with neuromuscular principles. One question haunted me for many years: is there a limit to how far the mandible can be advanced?

With the use of TENS, and the assistance of cross-checking with speech, I finally arrived at an answer. Clarity came slowly over the years, with sporadic insights from many cases. I have distilled my understanding into what I call the **Biomechanical Principles of Occlusion**.

The hypothesis

Biomechanical principles of all joints must also be satisfied for the temporomandibular joint.

From this proposition, several conditions follow:

- bullet** Hypermobility of joints has negative sequelae; joints function best when their range of motion is minimized.
- bullet** Excess range of motion increases stress on the musculature, joint tissues, and the supporting nervous system.
- bullet** The TMJ functions best when occlusion supports jaw functions within a minimal range of motion — i.e., when speech, rest, and open–close arcs are superimposed.
- bullet** Even though the TMJ has the ability to translate, it functions best when function requires it to translate least.

Corollaries

Satisfying these conditions implies the following clinical corollaries:

- bullet** Speech and centric occlusion should lie on the same trajectory.
- bullet** There should exist a fair arc in open–close without anterior or posterior shifting of the mandible.
- bullet** An end-on-end bite (Class III) creates an ideal functional jaw relationship in most cases — i.e., no translation is required for incising.

Validation

Validation of this hypothesis has come through many years of applying it to thousands of TMJ and orthodontic cases. The end-on-end bite is supported by the anthropological observation that all recovered primitive human skulls have an end-on-end bite. Dr. Percy Begg, the noted Australian orthodontist, arrived at the same conclusion — that the end-on-end bite is the most physiologic.

Clinical implications

Creating and maintaining an occlusion that satisfies these underlying orthopedic principles has required the development of treatment protocols that vary from those advocated by traditional orthodontic and

TMJ techniques. The first premise I had to abandon was the neuromuscular notion that TENS alone is an effective means of determining the initial cranio-mandibular relationship. European orthopedic concepts have shown that mandibular rest position can in fact be changed with appropriate appliances. The question then arises: *where should that position be created?*

Clinical experience from treating thousands of brittle TMJ cases has conclusively shown that all my patients do best when their mandible is positioned on the fair arc of open–close–speech. With that goal in mind, I begin treatment with orthopedic appliances. These appliances are fabricated to proprioceptively guide the patient toward an anteriorized position, which patients quickly accommodate to. That position is often on the fair arc of open–close–speech — and is frequently anterior to the position generated by TENS.

Anatomical splints are not always adequate to proprioceptively guide the mandible to the appropriate anteriorized trajectory. In those cases, I frequently use twin-block Crozat appliances. TENS is then used to further refine the bite once habituation to the appropriate trajectory has been achieved.

This process — though slower and more involved than conventional approaches — has proved necessary for effective resolution of the many associated medical complaints found in temporomandibular dysfunction. Many of my patients, from long-term trigeminal hypertonicity, have developed a generalized hypersensitivity syndrome that only begins to subside when they are precisely maintained on this trajectory. I believe this phenomenon is related to mesencephalic trigeminal influence on neuropeptides.

A fair arc of closure is often not generated with TENS, as multiple neuromuscular practitioners have noted (commonly seen as scan-1 imperfections). This appears to be due to a number of factors affecting resting mandibular position, including contracture, proprioceptive influences, joint compression, and body posture.

About the author. Dwight Jennings, DDS, MICCMO, has practiced jaw orthopedics and craniomandibular medicine in Alameda, California since 1986. He holds the Master designation (MICCMO) of the International College of Cranio-Mandibular Orthopedics and is the author of papers on the substance-P cascade, the trigeminal factor in systemic illness, and the cranio-mandibular contribution to Parkinson's, fibromyalgia, epilepsy, and autism-spectrum disorders.

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